HailGuard AI: Closed-Loop Self-Maintenance Healing Telemetry
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THINK. BUILD. CREATE. WIN

End-to-End System Architecture: Closed-Loop Fleet Maintenance

1. Data Engineering

- NASA C-MAPSS Telemetry Ingestion (21 Sensors + 3 Settings).
- Strict causality: Group-wise rolling means (w=5, 20) with no future lookahead.
- Engine ID isolation: 80/20 train/validation disjoint split.
- Zero-variance sensor filtering (sensors 1, 5, 10, 16, 18, 19 in FD001).

2. Predictive ML Engine

- Dual-engine approach: Degradation Anomaly Detection + RUL Regression.
- Piecewise Linear RUL Target: Capped at 125 cycles (healthy vs wear phases).
- Benchmarked 4 Models: Ridge, Random Forest, HistGradientBoosting, LightGBM.
- Validation-only selection: Official NASA test set evaluated strictly once.

3. Non-LLM Triage Agent

- Lightweight, deterministic rule engine running with sub-millisecond latency.
- Risk matrix: Combines RUL prediction + anomaly persistence.
- Automated Severity Assignment: P1 Critical (RUL < 15), P2 High (15-30), P3 Medium (30-50).
- Generates CMMS work orders with duplicate suppression & audit logs.

4. MLOps & Continuous Loop

- Experiment tracking: MAE, RMSE, and NASA PHM08 asymmetric loss.
- Automated retraining trigger on fleet data distribution drift.
- Human-in-the-Loop exception handling for borderline confidence.
- Modular architecture ready for multi-regime scaling (FD002–FD004).

Data Engineering: Causal Integrity & Leakage Prevention

Causal Feature Engineering Principles

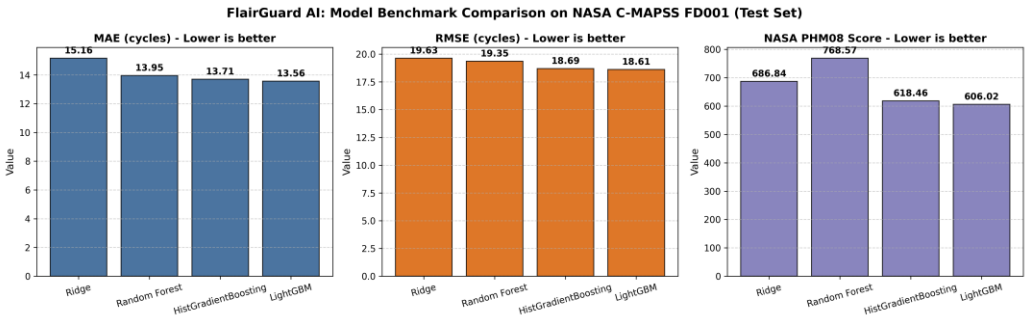
- Elimination of Lookahead Bias: In operational fleets, future cycles do not exist. Features at cycle 't' are computed strictly from historical cycles (1 to t) for that engine.
- Causal Rolling Statistics: Rolling window aggregations (w=5, 20) capture degradation acceleration without leaking future engine state.
- Group-Disjoint Splits: Standard random row splitting causes severe engine data leakage. FlairGuard groups strictly by engine_id (80 fit / 20 validation engines).
- Defensive Ingestion: Validates monotonic cycles, schema completeness, and verifies non-negative finite telemetry across all engines.

Physics-Informed Preprocessing

- Piecewise Linear Degradation: Real turbofans do not degrade linearly from day one. Wear initiates after initial cycles; RUL target is capped at 125 cycles during training.
- Regime & Sensor Invariance: Flat sensors with negligible variance in single-operating conditions are pruned to prevent ill-conditioned covariance matrices in linear models.
- Uncapped Official Testing: Test ground truth remains unclipped (evaluating up to 140+ cycles) to ensure honest, unadulterated NASA benchmark metrics.
- Automated Property Testing: 8 pytest unit tests enforce strict causality, engine isolation, and reproducible metrics.

Empirical ML Benchmarking on NASA C-MAPSS FD001

Model	FD001 MAE	FD001 RMSE	NASA PHM08 Score
Ridge (L2 Baseline)	15.16	19.63	686.84
Random Forest	13.95	19.35	768.57
HistGradientBoosting	13.71	18.69	618.46
LightGBM (Selected)	**13.56**	**18.61**	**606.02**



Benchmark Analysis & Model Selection

- LightGBM achieved the best overall performance with MAE of 13.56 cycles, RMSE of 18.61 cycles, and lowest NASA Score of 606.02.
- NASA PHM08 Metric: Heavily penalizes late predictions (predicting engine will survive longer than it does). LightGBM minimized catastrophic late failures.
- Zero Test Peeking: Selection was governed entirely by held-out validation engine scores; official test set was evaluated strictly once for final reporting.

Lightweight Non-LLM Triage Agent & Closed-Loop Actions

Deterministic Decision Logic (No LLM Required)

- Why Non-LLM? Telemetry decision loops require sub-second deterministic reliability, zero hallucination risk, and strict auditability under regulatory standards.
- Multi-Tier Severity Triage:
  - Critical (P1): RUL <= 15 cycles or persistent high-magnitude sensor anomaly. Generates immediate maintenance work order with expedited parts routing.
  - High (P2): 15 < RUL <= 30 cycles. Schedules priority servicing within 48 hours.
  - Medium (P3): 30 < RUL <= 50 cycles. Alerts fleet maintenance planner for routine scheduling.
  - Nominal (P4): RUL > 50 cycles. Continuous monitoring status.
- Work Order Deduplication: Prevents alert flooding by maintaining active work order hash states per engine.

Human-in-the-Loop & Auditability

- Automated CMMS Work Order Schema: Generates structured JSON/PDF work order containing engine ID, current cycle, predicted RUL, primary degrading sensor, and assigned technician role.
- Exception Escalation: When anomaly detector flags high divergence but regression predicts healthy RUL (model conflict), the agent escalates to senior reliability engineers for manual inspection.
- Audit Trail: Complete decision logs recorded with timestamp, model version, and exact input telemetry snapshot for full traceability.
- Closed-Loop Resolution: Fleet supervisor confirms or closes work order, providing ground truth feedback to the MLOps retraining pool.

MLOps Architecture & Multi-Regime Scaling Strategy

MLOps Pipeline & Continuous Retraining

- Model Registry & Versioning: Tracks candidate model weights, hyperparameters, and experiment artifacts across iterations.
- Drift Monitoring: Monitors Kolmogorov-Smirnov test statistics on incoming sensor streams against baseline distributions.
- Automated Retraining Trigger: Automatically initiates retraining pipeline when:
  - New run-to-failure engines reach maintenance completion.
  - Feature drift score exceeds critical tolerance thresholds.
  - Model validation error degrades past predetermined bounds.
- Production Serving: Lightweight serialized LightGBM artifacts deliver inference in under 5 milliseconds per telemetry snapshot.

Roadmap for FD002–FD004 (Multi-Regime Telemetry)

- FD001 vs Multi-Regime Challenge: FD001 has 1 operating condition and 1 fault mode. FD002 & FD004 have 6 operating conditions, causing sensor values to swing based on altitude/Mach rather than wear.
- Regime-Aware Normalization: K-Means operating condition clustering on settings (altitude, Mach, throttle) followed by cluster-specific standard scaling.
- Ensemble Scaling: Dual-mode degradations (HPC & fan degradation in FD003/FD004) handled via multi-head gradient boosting or temporal sequence models (GRU/LSTM).
- Interactive Monitoring UI: Streamlit fleet dashboard for real-time fleet health visibility and work-order management (Round 2 PoC).